

Prefix mincut

n_{stab} -compression of $R^\dagger$
is the frequently-cone; the floor is undecided

Stefan Hamann

August 30, 2026

Abstract

The r449 tail is not a Weil value. This note names the object the frequently-quantifier actually consumes: the r362/r369 border-augmented $R^\dagger$ cut at depth n_{stab} (not at $N_w - 3$, not at a flat prefix-80). On living selected windows the n_{stab} -delta and the full-window delta are *not* close (relative gap 0.97–3.62): two charts. The honest prefix chart (Selected $k \leq 9$ plus deep k_z 136, 137, 170, 197, 230, 500, N up to 17847) prefers M2; a positive floor is undecided. n_{stab} grows as $ca^{1.27}$. Dead χ remain dead on the *full* window; the prefix of χ lives (death in the last three grades). Lean identifies the named mincut with this prefix by `Iff.rfl`. No new `sorry`. No RH claim. No anti-RH claim.

Status. No new SATZ. Propositions 1–6 are a named infra census. Ausgang PREFIX_OBJECT_SPLIT / M2_UNDECIDED / NSTAB_GROWS / CHI_PREFIX_LIVES / LEAN_IDENT_RFL. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 Object

Proposition 1 (n_{stab} -compression). *Let n_{stab} be the r449 Chebyshev Fenster-Vergleich (skip m_0, m_1 ; first neighbour break; ε -robust). The prefix object is the r362/r369 construction with chain depth $n_\star = n_{\text{stab}}$ instead of $N_w - 3$:*

$$B_w = \left(\sum_{j=1}^{n_\star-1} \rho_j \right) + \frac{5}{7},$$

then $q^\dagger / \delta = 1 - q^\dagger$ from `solve.qdag` on that truncated chain plus the same border augmentation. Prefix-80 was a flat cut; this is the stabilized depth.

2 Living consistency

Proposition 2 (object split on living windows). *On every measured living window the full-window δ and the n_{stab} - δ disagree at relative gap ≥ 0.97 (k_z 5, 9, 17, 26, 43, 69, 116, 136). The prefix sits higher ($\delta \in [0.103, 0.214]$) than the full living chart ($\delta \in [0.028, 0.109]$). The Chebyshev cut is a moment-agreement depth, not the ABD-living depth: living windows keep accumulating C -resolvent past n_{stab} , so $q^\dagger$ rises and δ falls. This validates that the objects are distinct; it does not validate numerical closeness.*

3 Prefix δ table

k_z	N	n_{stab}	$q^\dagger$	δ	class
5	72	13	0.79918	0.20082	live / split
9	184	13	0.82998	0.17002	live / split
17	96	18	0.78568	0.21432	live / split
26	364	18	0.84063	0.15937	live / split
43	839	60	0.86133	0.13867	live / split
69	5690	119	0.89687	0.10313	live / split
116	1433	158	0.86848	0.13152	live / split
136	1641	175	0.87092	0.12908	live / split
137	8300	175	0.90011	0.09989	n_stab only
170	5771	194	0.89409	0.10591	n_stab only
197	4071	406	0.88786	0.11214	n_stab only
230	2012	194	0.87326	0.12674	n_stab only
500	17847	1525	0.90713	0.09287	n_stab only

All listed prefix packs have $n_{\text{flip}} = 0$ and pos_ok. Float engine (same as r445).

4 Floor fits

Proposition 3 (three charts, M2 on the prefix). *Three transparent fits (r421 diagnose_seq):*

1. **Slice only** (full-window $k = 5..9$): winner M1, $\delta_\infty = +0.02741$, $\text{AIC}_1 = -56.06 < \text{AIC}_2 = -50.95$. The r445 floor stands on its own chart.
2. **Mixed** (full-slice + n_{stab} -deep, fake $k = 10..15$, N up to 17847): winner M2, M1-grid hits -0.02 , $\text{AIC}_2 = -74.82 < \text{AIC}_1 = -72.92$. Drop-deep restores M1 ($\delta_\infty = 0.032$). Drop $k = 5$, drop $k = 9$, drop $k_z 500$: still M2.
3. **Object-pure** (every $n_{\text{stab}}\text{-}\delta$, abscissa a): winner M2, $\text{AIC}_2 = -101.17 < \text{AIC}_1 = -99.18$. Drop $k_z 17$ still M2 (M1 moves to $+0.061$); drop $k_z 500$ still M2; deep-only prefers M3.

The prefix-80 band 0.09–0.13 was not a higher floor: it was a flat shallow cut. The honest n_{stab} chart is higher than 0.027 but still decaying (M2). A positive prefix-floor is undecided.

5 n_{stab} scaling

Proposition 4 (n_{stab} grows). On the deep set $k_z 136, 137, 170, 197, 230, 500$,

$$n_{\text{stab}} \sim 0.0417 a^{1.273}.$$

Absolute depth grows ($175 \rightarrow 1525$ from $k_z 137 \rightarrow 500$). The ratio n_{stab}/N falls in $[0.021, 0.11]$: the cut is not a fixed fraction of the window. The extraction quantifier needs $n_{\text{stab}} \rightarrow \infty$ so that every finite $\text{meshExp}(f)$ is eventually inside the prefix after onset $a_0(f)$. Empirical growth supports that; it is not a theorem. Onset $a_0(f)$ remains finite per f .

6 Dead χ

Proposition 5 (prefix does not classify χ). *CHI₃-15 on the full window stays edge-dead: $q^\dagger = 1.023$, $n_{\text{flip}} = 0$. Its Chebyshev n_{stab} against neighbours 14/16 is 37. At $n = 40$ (and still at $n = 200$) the same r362/r369 pack is living ($q^\dagger = 0.863$ at $n = 40$). Death occurs in the last three grades ($200 \rightarrow 203$). The prefix object is the selected-window extraction cone, not a χ -classifier. Edge death remains a full-window statement.*

7 Lean naming

Proposition 6 (mincut is the prefix, Iff .rfl). *In $RH/FrequentlySelected.lean$, $frequently_selected_pre$ is definitionally $frequently_selected_augDualResolvent_ge_half$. $frequently_prefix_mincut_ident$ is Iff.rfl. The plain-read cone of the extraction ($fullRead$ for $meshExp(f) \leq m$ after onset) is the prefix; the ABD tail is not consumed. No new *sorry*. Census stays 5.*

8 Claim boundary

Finite-window census and a definitional Lean sharpening. Not a theorem of the Riemann hypothesis, nor evidence against it. No L^* claim. No $R^\dagger$ claim.